Phase 0 Diagnostic Test Results — ELF ASBM Coupling Analysis

Summary

All 8 tests from the **Phase 0 Diagnostic Test Plan** have been implemented and executed. The results provide a clear, evidence-backed recommendation.

> 🔴 **CAUTION**
> **Test 2 (the key decision-relevant test) returned NO-GO.** The OT-optimal coupling achieves only **2.11% reduction** in average squared path length vs. random pairing — well below the 5% minimum threshold. Per the plan's decision logic, this means the core premise of Idea 1 (non-memoryless coupling) does not hold at this scale.

Decision Log

Test	Result	Go / No-Go	Notes
1. Data prep	322K tokens, norms healthy	✅ N/A (setup)	Non-degenerate: norm_128 mean=2.89, std=0.41. PCA captures 66.5% variance in 128d.
2. OT vs. random coupling **(key test)**	**2.11% reduction**	❌ **NO-GO**	Far below 5% threshold. Random pairing is near-optimal.
3. Local curvature / intrinsic dim	Residual=0.097, ID=13.2	⚠️ MIXED	Low curvature but very low intrinsic dim (13/128). Manifold is flat but low-dimensional.
4. Token- vs. sequence-level structure	Token reduction 2.2% ≈ seq 2.1%	✅ GO	Coupling effect is consistent per-token (ratio 1.05×), but absolute effect is negligible.
5. ASBM compute scaling	24.8% overhead	❌ NO-GO	Scaling is $O(n^{2.09})$. Per-sequence coupling feasible but marginal at ELF batch size.
6. Prior energy sensitivity	22.8pp variation	❌ NO-GO	Prior choice drastically changes coupling on real embeddings (8%–31% reduction range).
7. Toy baseline reproduction	All 4 checks pass	✅ GO	L2↓, CE↓, no errors, non-degenerate sampling. Code is correct.
8. Toy coupling stability	Stable, CE decreases	✅ GO	No NaNs, no divergence, CE rate comparable to baseline.

Phase 0A — Embedding Geometry (Tests 1-4)

Test 1 — Data Preparation ✅

- **322,387 tokens** from 6,000 wikitext-2 sentences encoded with frozen T5-small
- 512-d → 128-d PCA projection (66.5% variance explained)
- Embedding norms: mean=2.89, std=0.41, range [1.21, 4.29] — non-degenerate
- Pairwise distances: mean=4.11, std=0.49 — healthy spread

Test 2 — OT vs. Random Coupling ❌ NO-GO

> ❗ **IMPORTANT**
> **This is the single most decision-relevant test.** Result: **2.11% reduction** (threshold was ≥15% for Go, <5% for No-Go).

- Random pairing cost: **137.44** (±0.15 over 50 shuffles)
- OT-optimal cost (Sinkhorn, $\varepsilon=0.05$): **134.53**
- **Ratio: 0.979** → only 2.1% improvement
- The independent/memoryless coupling leaves almost no transport cost on the table

Coupling comparison

Test 3 — Local Curvature ⚠️ MIXED

- Local PCA residual variance: mean=0.097 (low — manifold is locally flat)
- **TwoNN intrinsic dimensionality: 13.19** (much lower than 128-d bottleneck)
- PCA dims for 90/95/99% variance: 97/112/125
- **Interpretation**: The data lives on a ~13-d manifold but it's *locally linear*, meaning straight-line interpolation is near-optimal locally even though the data is low-dimensional

Curvature analysis

Test 4 — Token vs. Sequence ✅ (but moot)

- Token-level OT reduction: 2.2% (averaged across position groups)
- Sequence-level reduction: 2.1%
- **Ratio: 1.05×** → coupling effect is uniform across positions
- The effect doesn't evaporate per-token, but it's negligible at both levels

---

Phase 0B — ASBM Feasibility (Tests 5-6)

Test 5 — Compute Scaling ❌ NO-GO

- Scaling: time $\approx 1.7\times10^{-8} \cdot n^{2.09} \cdot d^{-0.22}$ (quadratic in sample size)
- Per-sequence coupling at ELF scale: **0.15ms** × 512 sequences = **0.074s/batch**
- Estimated forward pass: ~0.3s → **24.8% overhead**
- Just above the 20% threshold — feasible with approximation but not free

Scaling curves

Test 6 — Prior Sensitivity ❌ NO-GO

- On **real T5 embeddings**: reduction varies 8.2%–30.9% depending on prior choice
- **22.8 percentage-point variation** — prior selection is a first-class research question
- On synthetic data: 9.6pp variation (more moderate)
- The Gaussian prior (ELF default) gives the *weakest* coupling improvement

Prior sensitivity

Phase 0C — Training Pipeline (Tests 7-8)

Test 7 — Toy Baseline ✅ ALL PASS

- Model: 2-layer, 64-hidden, 125K params
- All 4 checks pass: L2↓ (20.0→13.0), CE↓ (5.60→5.55), no errors, non-degenerate sampling

Training curves

Test 8 — Coupling Stability ✅ GO

- OT-coupled training is stable (no NaNs, no divergence)
- L2 loss decreases similarly to baseline
- CE decode-branch loss decreases at comparable rate (ratio 0.990 vs baseline 0.991)
- Gradient norms are well-behaved

Stability curves

Overall Recommendation

Per the plan's decision logic:

> **"If Test 2 is No-Go, stop here regardless of other results — the core premise doesn't hold at this scale, and cluster time should go to the SDE/sampler project instead."**

❌ Do NOT proceed with the non-memoryless (ASBM-style) coupling project.

Why: At ELF's actual operating scale (T5-small embeddings, 128-d bottleneck), random pairing is already near-optimal (only 2.1% worse than OT-optimal). The manifold is locally flat (Test 3), so straight-line interpolation doesn't leave meaningful quality on the table. While the coupling *is* stable (Test 8) and the effect doesn't evaporate per-token (Test 4), the absolute effect is too small to justify the engineering effort.

Additional concerns even if Test 2 had been favorable: - Prior energy sensitivity (Test 6) would add a new unresolved hyperparameter - Compute overhead is borderline (Test 5: ~25% of forward pass)

Recommended next step

Redirect cluster time (Magus) to the **SDE/sampler project** instead, which doesn't depend on the same geometric assumption.

Files Created

File	Purpose
test_01_data_prep.py	Data prep & embedding extraction
test_02_ot_coupling.py	OT vs. random coupling (key test)
test_03_curvature.py	Local curvature & intrinsic dim
test_04_token_vs_seq.py	Token-level vs sequence-level
test_05_compute_scaling.py	ASBM compute scaling
test_06_prior_sensitivity.py	Prior energy sensitivity
test_07_toy_baseline.py	Toy baseline reproduction
test_08_coupling_stability.py	Coupling stability probe
phase0/results/	JSON results + PNG plots